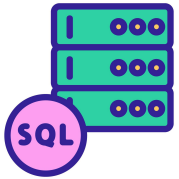
SQL
PART I





SQL Queries

01 Basic SELECT statement:

* **SELECT fields**
  FROM table
  WHERE condition
  ;

EXAMPLE DATABASE

```sql
CREATE TABLE student (
    studentId INTEGER PRIMARY KEY,
    studentName TEXT NOT NULL,
    studentStatus TEXT NOT NULL
);
```

student

studentId	studentName	studentStatus
1	Alice Zhang	Enrolled
2	Brian McCarthy	Enrolled
3	Carmen Lopez	Pending
4	David Kim	Graduated

* test repo in edstem

```sql
CREATE TABLE class (
    classId INTEGER NOT NULL,
    courseId INTEGER NOT NULL,
    classDescription TEXT NOT NULL,
    classDate TEXT NOT NULL,
    PRIMARY KEY (classId, courseId),
    FOREIGN KEY (courseId) REFERENCES course(courseId)
);
```

class

classId	courseId	classDescription	classDate
1	1	Week 1: SQL Basics	2025-09-01
2	1	Week 2: Joins	2025-09-08
1	2	Week 1: HTML	2025-09-01

```sql
CREATE TABLE enrollment (
    studentId INTEGER NOT NULL,
    courseId INTEGER NOT NULL,
    semester TEXT NOT NULL,
    PRIMARY KEY (studentId, courseId),
    FOREIGN KEY (studentId) REFERENCES student(studentId),
    FOREIGN KEY (courseId) REFERENCES course(courseId)
);
```

enrollment

studentId	courseId	semester
1	1	Fall 2025
1	2	Fall 2025
2	1	Fall 2025
3	3	Spring 2026
4	1	Spring 2025

```sql
CREATE TABLE program (
    programId INTEGER PRIMARY KEY,
    programName TEXT NOT NULL
);
```

program

programId	programName
1	Computer Science
2	Web Minor
3	Data Science Minor

course

courseId	courseName	instructorId	programId
1	Intro to Databases	1	1
2	Web Development	2	1
3	Python	3	2

```sql
CREATE TABLE course (
    courseId INTEGER PRIMARY KEY,
    courseName TEXT NOT NULL,
    instructorId INTEGER NOT NULL,
    programId INTEGER NOT NULL,
    FOREIGN KEY (instructorId) REFERENCES instructor(instructorId),
    FOREIGN KEY (programId) REFERENCES program(programId)
);
```

instructor

instructorId	instructorName	instructorEmail
1	Dr. Karlstein	karlstein@nyu.edu
2	Prof. Wong	wong@nyu.edu
3	Dr.Cordoza	chen@nyu.edu

```sql
CREATE TABLE instructor (
    instructorId INTEGER PRIMARY KEY,
    instructorName TEXT NOT NULL,
    instructorEmail TEXT NOT NULL
);
```

the WHERE clause can be simple or complex



- Conditions vary by data type
- Conditions can involve multiple fields and/or constants
- Conditions can be combined

Numeric Conditions (=,>,<, !=, >=, <=)

=	What is the status of the student with an id of 3?
sql	sqlite> SELECT studentId, studentName, studentStatus FROM student WHERE studentId = 3 ; 3|Carmen Lopez|Pending
>	What is the status of any student with an id greater than 3?
sql	sqlite> SELECT studentId, studentName, studentStatus FROM student WHERE studentId > 3 ; 4|David Kim|Graduated



String conditions – Exact match:

What students have graduated?

```
sqlite> SELECT studentId, studentName, studentStatus
   ...> FROM student
   ...> WHERE studentStatus = 'Graduated'
   ...> ;
4|David Kim|Graduated
```

String conditions - Wildcards

01

% - any number of characters

What is the status of students named John?

```
sqlite> SELECT studentId, studentName, studentStatus
   ...> FROM student
   ...> WHERE studentName LIKE 'Bri%'
   ...> ;
2|Brian McCarthy|Enrolled
```

02

_ only that character

What is the status of students with _I__e in their name?

```
sqlite> SELECT studentId, studentName, studentStatus
   ...> FROM student
   ...> WHERE studentName LIKE '_I__e%'
   ...> ;
1|Alice Zhang|Enrolled
```

03

% — characters before a pattern

What students have names that end in “im”?

```
sqlite> SELECT studentId, studentName, studentStatus
...> FROM student
   ...> WHERE studentName LIKE '%im';
4|David Kim|Graduated
```



Dates: 



SQLite has no dedicated date data type, but dates entered as TEXT in ISO format are still lexicographically sortable:

ISO date format =

YYYY-MM-DD

```
SELECT classDescription, classDate
FROM class
WHERE classDate BETWEEN '2021-01-01' AND '2021-12-31';
```

DateTime Conditions in SQLite

FORMATTING

Care must be taken in formatting dates

Use < and > to mean before or after

Use BETWEEN between two dates

Example 1

What classes happened during or after March 2021?

```
sqlite> SELECT classDescription, classDate
FROM class
WHERE classDate >= '2021-03-01'
;
Week 1: SQL Basics|2025-09-01
Week 2: Joins|2025-09-08
Week 1: HTML|2025-09-01
```

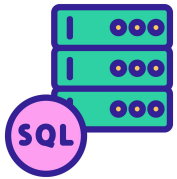
Example 2

What classes happened between September 1 - 7 2025?

```
sqlite> SELECT classDescription, classDate
FROM class
WHERE classDate BETWEEN '2025-09-01' AND '2025-09-07';
Week 1: SQL Basics|2025-09-01
Week 1: HTML|2025-09-01
```



COMBINING CONDITIONS W/ LOGICAL OPERATORS

&	AND	What classes happened before mid-September 2025 and are about SQL?	sqlite> SELECT classDescription, classDate FROM class WHERE classDate < '2025-09-15' AND classDescription LIKE '%SQL%'; Week 1: SQL Basics|2025-09-01
			
||	OR	What classes have an id of 1 or 3?	sqlite> SELECT classId, classDescription ...> FROM class ...> WHERE classId =1 OR classId = 2 ...> ; 1|Week 1: SQL Basics 1|Week 1: HTML 2|Week 2: Joins

ADDENDUM:

01

WHERE condition fields do not have to be in the SELECT list of fields

What classes happened after September 1, 2025?

```
sqlite> SELECT classDescription
   ...> FROM class
   ...> WHERE classDate >
'2025-09-01';
Week 2: Joins
```

02

What dates had classes about SQL?

```
sqlite> SELECT classDate
FROM class
WHERE classDescription LIKE '%SQL%';
2025-09-01
```

03

What students are currently enrolled?

```
sqlite> SELECT studentName
FROM student
WHERE studentStatus = 'Enrolled';
Alice Zhang
Brian McCarthy
```

1. What student IDs belong to students whose names contain “m”?

2. What classes happened after September 1, 2025 AND are not about HTML? HINT: ‘NOT LIKE’

3. What course names belong to programId 1 OR programId 2?

4. What instructor emails belong to instructors whose names start with “Dr.”?

5. What semesters include courseId 1?

— BONUS. What program names contain the word “Science”?



Challenges